

A2

Agreements

Both the DBSCAN and K-Means clusters agree on the downtown Vancouver area as being a dense business hub. This is shown through the big cluster at the top of the DBSCAN plot and green cluster at the top of the K-Means plot. Additionally, they both seem to follow the same map of Vancouver. This is recognizable as they both have the same distinct shape such as that diagonal line in the bottom left. However, this makes sense as they were both fed the same latitude and longitude coordinates.

Disagreements

Where they differ is that K-means breaks down Vancouver into 6 clusters, meaning it generalizes an area and forces every point into a cluster and region of Vancouver. On the other hand, DBSCAN breaks down Vancouver much more. To be more specific, it broke down Vancouver into 248 clusters, meaning it isolated a lot of areas. It then also identified businesses that did not seemingly fit into any smaller area and classified them as noise.

Assumptions

K-means assumes that all clusters are approximately the same size, hence, it tries breaking up the businesses into 6 (k value) equally-sized clusters. It lumps low-density business areas into high-density business-areas because it cannot handle a variation in density.

DBSCAN does not assume anything about the shape. It does not have a set amount of clusters it is trying to break the dataset into. Instead it uses its eps value and min_samples value to break down the area into clusters, with each cluster being within the 0.03 eps (eps value) of each other and each cluster having a minimum of 10 businesses (min_samples). The points that do not become a part of a core point due to not being within range of other clusters and not being within range of a minimum of 9 other businesses to create their own core point. Due to this, they are then classified as noise and counted under the -1 group.

A3

Result Representation

Feature-based clustering grouped together businesses that have similar characteristics such as number of employees, fees paid, duration of their license, amount of times their license has been revised, industry, and which month their license was issued. Each cluster seems to have a defining feature. For instance, cluster 4 seems to have very large businesses with an average

employee count above 1000. This makes sense as the primary industry in this cluster is financial services. Cluster 0 seems to be full of more medium-sized businesses with moderate fees and long licence durations and mostly grouped into the "Other" industry category. Cluster 1 consists of smaller businesses with lower fees and slightly shorter licence durations. Cluster 2 consists of businesses with the shortest licence durations and lowest fees and employee count which suggests that this is the group of the newest businesses that have the smallest operations. Cluster 3 has a substantially higher license revision count in comparison to other clusters meaning that they likely get their licenses renewed or fixed a lot. Cluster 5 seems to consist of big businesses as well as they have extremely high licence fees, are primarily long-term rental businesses, and have the second-highest average employee count. Cluster 6 is similar to cluster 0 as they both seem like medium-sized businesses but cluster 6 seems slightly smaller as it has lower average fees, number of employees, and license duration. Lastly, cluster 7 seems to be an industry-specific cluster similar to cluster 4 and 5. Cluster 7 caters to health care professionals and services.

Difference from A2 Clustering

Yes this does differ from the geographic clustering in A2. In A2, businesses were grouped together and formed clusters based on neighbourhoods and their business density regardless of their characteristics. In contrast, A3 groups together businesses that share similar features and characteristics regardless of their geographic location within Vancouver. This means that businesses from different areas, even opposite sides of the city, can belong to the same feature-based cluster if they have a similar employee count, amount of fees paid, license duration, license revision number, industry, and license issued month. Whereas A2 clustering caters toward location and geography, A3 clustering caters towards individual business features.

B1

For B1, I decided to clean the original dataset by keeping only issued licenses and removing business records with no local area and no business type because these fields are needed to make an area-level feature vector. I did not want to impute these values because I did not want to introduce artificial data using the mode or other imputation strategies that could possibly alter or distort the results.

Additionally, I decided to utilize local areas because it is cleaner and does not have as many missing values as Postal FSA which means I retain more data records.

I decided to exclude areas with less than 50 businesses because the sample size would be quite small. In this case, a few businesses would heavily influence that area's business type count meaning everything would be amplified.

B4

For B4, I explored multiple K values, however, I believe K=5 is the more optimal K value to observe. This is because anything above results in more clusters where there is simply one area. On the other hand, anything below results in areas seemingly being lumped and forced together. K=5 strikes a good balance between the two by acknowledging the individuality of each area but also recognizing similar characteristics.

Cluster 0 (Victoria-Fraserview, Hastings-Sunrise, Sunset, Killarney, and Renfrew-Collingwood):

This cluster mainly includes east and southeast Vancouver areas. These areas are generally considered residential and consist of smaller businesses and local shopping streets. Therefore they were likely grouped together due to the fact they are neighbourhood and local-oriented business areas.

Cluster 1 (Out of Town):

It makes sense that the "Out of Town" category forms its own cluster. This is because it does not represent a Vancouver area/neighbourhood. Instead, this category consists of businesses located out of Vancouver's local areas and hence, it has a different business type composition from other Vancouver neighbourhoods.

Cluster 2 (Downtown, Fairview, and Strathcona):

This cluster mainly consists of neighbourhoods full of commercial centers. Downtown has the highest number and concentration of businesses, and Fairview and Strathcona are somewhat similar and also have high commercial activity, restaurants, and other retail businesses. Therefore, due to their similar business type composition, it makes sense they are grouped together.

Cluster 3 (Dunbar-Southlands, South Cambie, Shaughnessy, and Oakridge):

This cluster consists of neighbourhoods and areas that are considered more residential. Therefore, they have lower commercial and business density than other neighbourhoods such as Downtown. Overall, these neighbourhoods likely have more businesses focused on locality and neighborhood services resulting in similar business type distributions.

Cluster 4 (Arbutus Ridge, Riley Park, Mount Pleasant, Marpole, Kitsilano, Kerrisdale, Kensington-Cedar Cottage, Grandview-Woodland, West End, and West Point Grey):

This cluster is the most populated. Although this may seem concerning given that these neighbourhoods are located throughout Vancouver and the likelihood of them being grouped together must mean that the K value is too low, I did experiment and a majority of these neighbourhoods stick together in a cluster even as the K value increases. This means that they share a similar business type distribution overall.